

Volume 3 — Mesa Conversion & Installation

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

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Table of Contents

- 1. Architecture Decision Summary
- 2. Hardware Stack & Bill of Boards
- 3. Control PC & Ethernet Setup
- 4. Mesa 7i97T — Primary Motion Controller
- 5. Mesa 7i49 — Resolver Feedback Interface
- 6. Mesa 7i84U — Remote Field I/O
- 7. LinuxCNC Configuration Files
- 8. Bring-Up Procedure (8 Stages)
- 9. Verification Checklists
- 10. Rejected / Historical Architecture

Chapter 1 — Architecture Decision Summary

Selected: LinuxCNC PC → Ethernet (`hm2_eth` , static IP) → **Mesa 7i97T** analog servo controller, with a **Mesa 7i49** (plain) resolver interface for axis feedback and a **Mesa 7i84U** smart-serial field-I/O card near the existing green breakout PCB.

Why it fits this machine:

- Three analog servo axes + analog spindle command match the 7i97T exactly.
- The machine keeps its original **Tamagawa resolvers** — so feedback is resolver-to-digital (7i49), not quadrature encoder.
- Ethernet removes dependence on a PCIe slot and lets the PC sit in the old Mazatrol rack area.
- The 7i84U mounts next to the existing terminal strips, keeping ATC/hydraulic field wiring short.
- A **sister retrofit** (Mazak VQC 15/40, same Mitsubishi TRA drives and HD81-12S motors, plain 7i49 at 5 kHz) validates the approach — see `srdco/MazakVQC1540` on GitHub.

Chapter 2 — Hardware Stack & Bill of Boards

Item	Role	Notes
LinuxCNC control PC	Trajectory, GUI, HAL	Must pass latency test before install
Mesa 7i97T	X/Y/Z ±10 V analog commands, drive enables, spindle analog, limits/homes/E-stop, SSR outputs	Ethernet, static IP 192.168.1.121
Mesa 7i49 (plain)	Resolver-to-digital, X/Y/Z channels RES0–RES2	5 kHz excitation baseline; 7i49HV is contingency only
Mesa 7i84U	32 in / 16 out remote field I/O	Smart-serial from 7i97T TB4

Interposing relays RLY-1...7	Drive 100 VAC solenoid coils	With flyback / RC suppression
WHB04B pendant (optional)	MPG jogging	USB to PC, install last

Do not order a second smart-serial card unless the input count in `current_pin_authority.csv` is proven insufficient (project rule).

Chapter 3 — Control PC & Ethernet Setup

1. Install LinuxCNC (current stable) on the PC; run the **latency test** for at least an hour with typical desktop load. Servo-thread jitter must be acceptable before proceeding.
2. Configure the NIC on the Mesa subnet with a static host address (e.g. 192.168.1.1/24). No DHCP on this link.
3. Verify: `ping 192.168.1.121`, then `mesaflash --device 7i97t --addr 192.168.1.121 --readhmid`.
4. **Save the outputs** as `mesa_readhmid.txt` and, after firmware load, the HAL pin dump as `mesa_hal_pins.txt` — these replace every placeholder pin name in the HAL files.

Chapter 4 — Mesa 7i97T — Primary Motion Controller

Connector	Function on this machine
TB3	X/Z/Y analog velocity commands (AOUT0/1/2) + drive S-ON enables (ENA0/1/2); spindle speed AOUT3
TB4	Smart-serial link to 7i84U (absolute pins TB4.13–TB4.20)
TB5	Limits (IN0–IN5), homes (IN6–IN8), E-stop monitor (IN9), 24 V field power, SSR outputs 0–5
TB2	Axis encoder inputs NOT USED (resolver feedback); encoder.03 reserved for a machine spindle encoder if fitted

Confirm the correct firmware/bitfile exposes: 3+ analog channels, smart-serial on TB4, and the required GPIO. Note whether the board tag is `hm2_7i97t` or `hm2_7i97` in the real pin dump.

Axis-to-channel order is X=0, Z=1, Y=2 on TB3 (matches pin authority) — do not assume X/Y/Z order.

Chapter 5 — Mesa 7i49 — Resolver Feedback Interface

The most machine-specific part of this conversion. Read Vol 4 §3 alongside this chapter.

Facts & baseline:

- Resolvers: Tamagawa (RT-5XA-11 / RT-5XA-L1 documented; per-axis labels still to confirm). MS3108B 20–29P is a **connector shell**, not a resolver model.
- These are ~2:1 control-transformer resolvers; the 7i49 drives ~2 V RMS and expects ~1 V RMS sin/cos back — a natural match, hence **plain 7i49**.
- Excitation: **5 kHz** (spec 4.5 kHz; 7i49 offers 2.5/5/10). Firmware `num_resolvers` must expose 3 channels.

Mandatory verification sequence (drives disabled):

1. **Ohmmeter the winding pairs per axis before any power.** Original Meldas/TRA wiring may run the resolver backwards (excitation into stator). Expected once verified: rotor pair (likely RO1/RO2) → RESDRV+/-; two matched stator pairs → RESSIN and RESCOS.
2. Confirm the **7i49 is the sole excitation source** — nothing from the old drive/control may still touch the windings. (TRA drives close their velocity loop on the tachometer, so they don't need the resolver.)
3. Energize, **scope the sin/cos return** (~1 V RMS expected). Low → noise/sluggish response. Too hot → divider or W2 half-drive jumper (**field option, not default**). Far too weak at full drive → only then consider 7i49HV.
4. Rotate each axis by hand; confirm counts, direction, and derive the resolver scale (counts per machine unit) for the INI.

Chapter 6 — Mesa 7i84U — Remote Field I/O

- Mounts near the green breakout PCB; connects to 7i97T TB4 over smart-serial (full link map in Vol 4 §5). Shield drain lands at the **7i97T end only**.
- Field power banks: TB1 pins 1/2 (bank A → outputs 0–7, inputs 0–15), pins 3/4 (bank B → outputs 8–15, inputs 16–31). Confirm capacity and fusing from the HR-11F-24 24 V system before landing.
- Carries all ATC, magazine, hydraulic, coolant, lube, spindle-drive digital, panel, and alarm I/O — complete maps in Vol 4 §§8–9.
- Outputs that drive 100 VAC solenoid coils go through interposing relays RLY-1...RLY-7, never directly.

Chapter 7 — LinuxCNC Configuration Files

File	Purpose
mazak_vqc_20_40.ini	Machine, joints, spindle, HAL loading
mazak_vqc_20_40.hal	Main loader — loadrt hm2_eth board_ip=192.168.1.121
motion_7i97t.hal	Axis commands, resolver feedback nets, limits/homes/E-stop, enables
field_7i84u.hal	ATC, magazine, coolant, lube, hydraulic, panel I/O
postgui.hal, pendant_whb04b.hal	GUI extras, optional pendant
tool.tbl	Tool table — must match magazine pockets

These files are bring-up skeletons. Every `hm2_7i97t.*`/7i84U pin name, resolver scale, analog polarity, and NO/NC assumption is a placeholder until replaced with values from the real pin dump and cabinet measurements. Known stale nets are listed in the pin authority cleanup notes (e.g. tool clamp/unclamp confirm inputs moved from IN0/IN1 to IN15/IN16).

Chapter 8 — Bring-Up Procedure (8 Stages)

1. **Comms:** static IP, ping, mesaflash, `hm2_eth` loads, save readhmid + HAL pin dumps.
2. **24 V system:** trace HR-11F-24 (+S/+/-/-S, TOG, CNT, FG), P24/G24 distribution, remote sense, branch fusing, 0 V reference strategy.

3. **Resolvers (drives disabled):** full Chapter 5 sequence — ohmmeter, sole-excitation check, 5 kHz, scope, counts/direction/scale.
4. **Analog commands (drives inhibited):** verify 0 V at zero command; verify polarity per axis. A *reversed command* = runaway.
5. **One axis at a time:** low gain, low speed, small moves; tune; repeat per axis.
6. **Homes & limits:** prove switch logic before first homing cycle.
7. **Spindle:** verify FR-SX command mode (0–10 V vs bipolar — unresolved, see below), scaling at low RPM, run/direction/enable, zero-speed, at-speed, gear shifts, orient.
8. **ATC/hydraulics:** one output at a time, dry-run interlocks, no tool load.

Open decision to resolve during stage 7: spindle run/enable ownership — TB3 ENA3 candidate **conflicts** with the 7i84U FWD/REV/ENA plan (status HOLD_CONFLICT). Resolve before wiring either.

Chapter 9 — Verification Checklists

Pre-power: cabinet photos captured (checklist in repo `docs/`); safety chain traced and drawn; drive model/terminal labels recorded; FR-SX terminals recorded; resolver winding pairs ohm'd; 24 V fusing confirmed; nothing else on resolver windings.

Pre-motion: E-stop chain functional test (every button, door, overtravel final drops the contactors); estop-ext seen by LinuxCNC; drive enables verified per axis; zero-command verified; resolver counts/direction verified; following-error limits set conservatively.

Pre-ATC: every prox normal state measured (PRS-8/9/10/12/13/21–25, PHS-181/182, PRS-55/66); every solenoid coil voltage/current measured; relay/suppression fitted; hydraulic pressure switch proven; magazine BCD decode verified against physical pockets; dry-run with no tools.

Chapter 10 — Rejected / Historical Architecture

An earlier plan used a tower PC with a **6i25 PCIe** card, **7i77** analog servo card, **7i84** field I/O, and optional 7i85/7i85S. It was **superseded** by the 7i97T + 7i84U Ethernet architecture. Any document or workbook tab referencing 7i77 terminals or Magnescale quadrature encoders is **legacy** — do not wire from it. (Legacy tabs are clearly separated in the wiring master workbook.)

End of Volume 3. Continue to Volume 4 — Wiring Diagrams & I/O Reference.